

Data Analysis with Stata 15 Cheat Sheet

For more info see Stata's reference manual (stata.com)
Results are stored as either **r**-class or **e**-class. See Programming Cheat Sheet
Summarize Data Examples use auto.dta (sysuse auto, clear) unless otherwise noted

univar price mpg, boxplot ssc install univar
calculate univariate summary, with box-and-whiskers plot

stem mpg
return stem-and-leaf display of mpg

summarize price mpg, detail frequently used commands are highlighted in yellow
calculate a variety of univariate summary statistics

ci mean mpg price, level(99) for Stata 13: ci mpg price, level(99)
compute standard errors and confidence intervals

correlate mpg price
return correlation or covariance matrix

pwcorr price mpg weight, star(0.05)
return all pairwise correlation coefficients with sig. levels

mean price mpg
estimates of means, including standard errors

proportion rep78 foreign
estimates of proportions, including standard errors for categories identified in varlist

ratio
estimates of ratio, including standard errors

total price
estimates of totals, including standard errors

Statistical Tests

tabulate foreign rep78, chi2 exact expected
tabulate foreign and repair record and return chi² and Fisher's exact statistic alongside the expected values

ttest mpg, by(foreign)
estimate t test on equality of means for mpg by foreign

prtest foreign == 0.5
one-sample test of proportions

ksmirnov mpg, by(foreign) exact
Kolmogorov-Smirnov equality-of-distributions test

ranksum mpg, by(foreign)
equality tests on unmatched data (independent samples)

anova systolic drug webuse systolic, clear
analysis of variance and covariance

pwmean mpg, over(rep78) peffects mcompare(tukey)
estimate pairwise comparisons of means with equal variances include multiple comparison adjustment

Estimation with Categorical & Factor Variables

CONTINUOUS VARIABLES	OPERATOR	DESCRIPTION	EXAMPLE
measure something	i.	specify indicators	regress price i.rep78
CATEGORICAL VARIABLES identify a group to which an observations belongs	ib.	specify base indicator	regress price ib(3).rep78
	fvset	command to change base	fvset base frequent rep78
	c.	treat variable as continuous	regress price i.foreign#c.mpg i.foreign
INDICATOR VARIABLES T F denote whether something is true or false	o.	omit a variable or indicator	regress price io(2).rep78
	#	specify interactions	regress price mpg#c.mpg#c.mpg
	##	specify factorial interactions	regress price c.mpg##c.mpg

Declare Data

By declaring data type, you enable Stata to apply data munging and analysis functions specific to certain data types

TIME SERIES

webuse sunspot, clear

tsset time, yearly
declare sunspot data to be yearly time series

tsreport
report time series aspects of a dataset

generate lag_spot = L1.spot
create a new variable of annual lags of sun spots

tsline spot
plot time series of sunspots

arma spot, ar(1/2)
estimate an auto-regressive model with 2 lags

TIME SERIES OPERATORS

L. lag x_{t-1}	L2. 2-period lag x_{t-2}
F. lead x_{t+1}	F2. 2-period lead x_{t+2}
D. difference $x_t - x_{t-1}$	D2. difference of difference $x_t - x_{t-1} - (x_{t-1} - x_{t-2})$
S. seasonal difference $x_t - x_{t-12}$	S2. lag-2 (seasonal difference) $x_t - x_{t-2}$

USEFUL ADD-INS

tscollapse compact time series into means, sums and end-of-period values

carryforward carry non-missing values forward from one obs. to the next

tsspell identify spells or runs in time series

SURVIVAL ANALYSIS

webuse drugtr, clear

stset studytime, failure(died)
declare survey design for a dataset

stsum
summarize survival-time data

stcox drug age
estimate a Cox proportional hazard model

1 Estimate Models

stores results as **e**-class

regress price mpg weight, vce(robust)
estimate ordinary least squares (OLS) model on mpg weight and foreign, apply robust standard errors

regress price mpg weight if foreign == 0, vce(cluster rep78)
regress price only on domestic cars, cluster standard errors

reg price mpg weight, genwt(reg_wt)
estimate robust regression to eliminate outliers

probit foreign turn price, vce(robust)
estimate probit regression with robust standard errors

logit foreign headroom mpg, or
estimate logistic regression and report odds ratios

bootstrap, reps(100): regress mpg /*
*/ weight gear foreign
estimate regression with bootstrapping

jackknife r(mean), double: sum mpg
jackknife standard error of sample mean

ADDITIONAL MODELS

pca ← built-in Stata command principal components analysis

factor user-written factor analysis

poisson • **nbrreg** count outcomes

tobit censored data

ivregress • **ivreg2** instrumental variables

diff user-written difference-in-difference

rd ssc install ivreg2 regression discontinuity

xtabond • **xtpdpsys** dynamic panel estimator

teffects psmatch propensity score matching

synth synthetic control analysis

oaxaca Blinder-Oaxaca decomposition

more details at <http://www.stata.com/manuals/u25.pdf>

PANEL / LONGITUDINAL

webuse nlswork, clear

xtset id year
declare national longitudinal data to be a panel

xtdescribe
report panel aspects of a dataset

xtsum hours
summarize hours worked, decomposing standard deviation into between and within components

xtline ln_wage if id <= 22, tlabel(#3)
plot panel data as a line plot

xtreg ln_w c.age##c.age ttl_exp, fe vce(robust)
estimate a fixed-effects model with robust standard errors

SURVEY DATA webuse nhanes2b, clear

svyset psuid [pweight = finalwgt], strata(stratid)
declare survey design for a dataset

svydescribe
report survey data details

svy: mean age, over(sex)
estimate a population mean for each subpopulation

svy, subpop(rural): mean age
estimate a population mean for rural areas

svy: tabulate sex heartatk
report two-way table with tests of independence

svy: reg zinc c.age##c.age female weight rural
estimate a regression using survey weights

2 Diagnostics

some are inappropriate with robust SEs

estat hettest test for heteroskedasticity

ovtest test for omitted variable bias

vif report variance inflation factor

dfbeta(length) calculate measure of influence

nvfplot, yline(0) plot residuals against fitted values

avplots plot all partial-regression leverage plots in one graph

Type help regress postestimation plots for additional diagnostic plots

3 Postestimation

commands that use a fitted model

regress price headroom length Used in all postestimation examples

display _b[length] **display _se[length]**
return coefficient estimate or standard error for mpg from most recent regression model

margins, dydx(length) returns e-class information when post option is used
return the estimated marginal effect for mpg

margins, eyex(length)
return the estimated elasticity for price

predict yhat if e(sample)
create predictions for sample on which model was fit

predict double resid, residuals
calculate residuals based on last fit model

test headroom = 0
test linear hypotheses that headroom estimate equals zero

lincom headroom - length
test linear combination of estimates (headroom = length)

Data Processing

with Stata 15 Cheat Sheet

For more info see Stata's reference manual (stata.com)

Useful Shortcuts

- F2** — keyboard buttons describe data
Ctrl + 9 open a new .do file
Ctrl + 8 open the data editor
Ctrl + D highlight text in .do file, then ctrl + d executes it in the command line
clear delete data in memory

AT COMMAND PROMPT

- PgUp** **PgDn** scroll through previous commands
Tab autocompletes variable name after typing part
cls clear the console (where results are displayed)

Set up

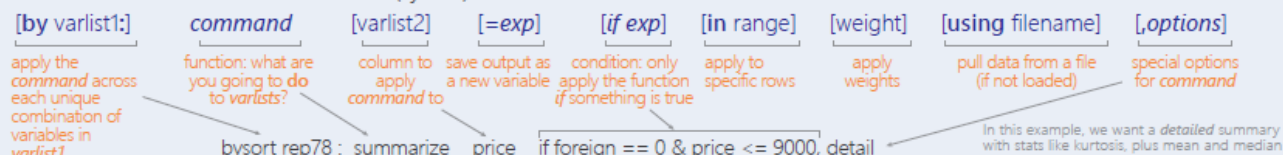
- pwd** print current (working) directory
cd "C:\Program Files (x86)\Stata13" change working directory
dir display filenames in working directory
dir *.dta List all Stata data in working directory
capture log close close the log on any existing do files
log using "myDoFile.txt", replace create a new log file to record your work and results
search mdesc find the package mdesc to install
ssc install mdesc install the package mdesc; needs to be done once
- underlined parts are shortcuts — use "capture" or "cap"*
packages contain extra commands that expand Stata's toolkit

Import Data

- sysuse auto, clear** load system data (Auto data)
use "yourStataFile.dta", clear load a dataset from the current directory
import excel "yourSpreadsheet.xlsx", /*
***/ sheet("Sheet1") cellrange(A2:H11) firstrow** import an Excel spreadsheet
import delimited "yourFile.csv", /*
***/ rowrange(2:11) colrange(1:8) varnames(2)** import a .csv file
webuse set "https://github.com/GeoCenter/StataTraining/raw/master/Day2/Data"
webuse "wb_indicators_long" set web-based directory and load data from the web
- for many examples, we use the auto dataset.*
frequently used commands are highlighted in yellow

Basic Syntax

All Stata commands have the same format (syntax):



To find out more about any command — like what options it takes — type **help command**

Basic Data Operations

Arithmetic
 + add (numbers)
 + combine (strings)
 - subtract
 * multiply
 / divide
 ^ raise to a power

Logic
 & and
 ! or ~ not
 | or
 == tests if something is equal
 < less than
 <= less than or equal to
 > greater than
 >= greater or equal to

if foreign != 1 & price >= 10000 if foreign != 1 | price >= 10000

make	foreign	price
Chevy Colt	0	3,984
Buick Riviera	0	10,372
Honda Civic	1	4,499
Volvo 260	1	11,995

Change Data Types

Stata has 6 data types, and data can also be missing:
 no data missing true/false byte words string int long float double

To convert between numbers & strings:

```

gen foreignString = string(foreign)
tostring foreign, gen(foreignString)
decode foreign, gen(foreignString)

gen foreignNumeric = real(foreignString)
destring foreignString, gen(foreignNumeric)
encode foreignString, gen(foreignNumeric)
    
```

recast double mpg
 generic way to convert between types

Explore Data

VIEW DATA ORGANIZATION
describe make price display variable type, format, and any value/variable labels
count
count if price > 5000 number of rows (observations)
 Can be combined with logic
ds, has(type string)
lookfor "in." search for variable types, variable name, or variable label
isid mpg check if mpg uniquely identifies the data

SEE DATA DISTRIBUTION
codebook make price overview of variable type, stats, number of missing/unique values
summarize make price mpg print summary statistics (mean, stdev, min, max) for variables
inspect mpg show histogram of data, number of missing or zero observations
histogram mpg, frequency plot a histogram of the distribution of a variable

BROWSE OBSERVATIONS WITHIN THE DATA

browse or **Ctrl + 8** Missing values are treated as the largest positive number. To exclude missing values, ask whether the value is less than .
 open the data editor
list make price if price > 10000 & !missing(price) **clist ...** (compact form)
 list the make and price for observations with price > \$10,000
display price[4] display the 4th observation in price; only works on single values
gsort price mpg (ascending) **gsort -price -mpg** (descending)
 sort in order, first by price then miles per gallon
duplicates report finds all duplicate values in each variable
assert price!=. verify truth of claim
levelsof rep78 display the unique values for rep78

Summarize Data

include missing values create binary variable for every rep78 value in a new variable, repairRecord

tabulate rep78, mi gen(repairRecord)
 one-way table: number of rows with each value of rep78
tabulate rep78 foreign, mi
 two-way table: cross-tabulate number of observations for each combination of rep78 and foreign
bysort rep78: tabulate foreign
 for each value of rep78, apply the command tabulate foreign
tabstat price weight mpg, by(foreign) stat(mean sd n)
 create compact table of summary statistics
 formats numbers for all data
table foreign, contents(mean price sd price) f(%9.2fc) row
 create a flexible table of summary statistics
collapse (mean) price (max) mpg, by(foreign) — replaces data
 calculate mean price & max mpg by car type (foreign)

Create New Variables

generate mpgSq = mpg^2 **gen byte lowPr = price < 4000**
 create a new variable. Useful also for creating binary variables based on a condition (generate byte)
generate id = _n **bysort rep78: gen repairIdx = _n**
 _n creates a running index of observations in a group
generate totRows = _N **bysort rep78: gen repairTot = _N**
 _N creates a running count of the total observations per group
pctile mpgQuartile = mpg, nq = 4
 create quartiles of the mpg data
egen meanPrice = mean(price), by(foreign) see help egen for more options
 calculate mean price for each group in foreign

Data Transformation with Stata 15 Cheat Sheet

For more info see Stata's reference manual ([stata.com](http://www.stata.com))

Select Parts of Data (Subsetting)

SELECT SPECIFIC COLUMNS

- drop make**
remove the 'make' variable
- keep make price**
opposite of drop; keep only variables 'make' and 'price'

FILTER SPECIFIC ROWS

- drop if mpg < 20**
drop observations based on a condition (left)
- drop in 1/4**
or rows 1-4 (right)
- keep in 1/30**
opposite of drop; keep only rows 1-30
- keep if inrange(price, 5000, 10000)**
keep values of price between \$5,000 – \$10,000 (inclusive)
- keep if inlist(make, "Honda Accord", "Honda Civic", "Subaru")**
keep the specified values of make
- sample 25**
sample 25% of the observations in the dataset
(use **set seed #** command for reproducible sampling)

Replace Parts of Data

CHANGE COLUMN NAMES

- rename (rep78 foreign) (repairRecord carType)**
rename one or multiple variables

CHANGE ROW VALUES

- replace price = 5000 if price < 5000**
replace all values of price that are less than \$5,000 with 5000
- recode price (0 / 5000 = 5000)**
change all prices less than 5000 to be \$5,000
- recode foreign (0 = 2 "US") (1 = 1 "Not US"), gen(foreign2)**
change the values and value labels then store in a new variable, foreign2

REPLACE MISSING VALUES

- mvdecode _all, mv(9999)** *useful for cleaning survey datasets*
replace the number 9999 with missing value in all variables
- mvencode _all, mv(9999)** *useful for exporting data*
replace missing values with the number 9999 for all variables

Label Data

Value labels map string descriptions to numbers. They allow the underlying data to be numeric (making logical tests simpler) while also connecting the values to human-understandable text.

label define myLabel 0 "US" 1 "Not US"

label values foreign myLabel

define a label and apply it to the values in foreign

label list

list all labels within the dataset

note: data note here

place note in dataset

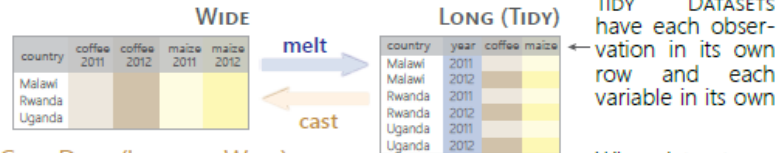
Reshape Data

webuse set [https://github.com/GeoCenter/StataTraining/raw/master/Day2/Data/webuse "coffeeMaize2.dta"](https://github.com/GeoCenter/StataTraining/raw/master/Day2/Data/webuse%20coffeeMaize2.dta) *load demo dataset*

MELT DATA (WIDE → LONG)

reshape variables starting with coffee and maize
unique id variable (key)
create new variable which captures the info in the column names

reshape long coffee@ maize@, i(country) j(year) *new variable*
convert a wide dataset to long



CAST DATA (LONG → WIDE)

create new variables named coffee2011, maize2012...
what will be unique id variable (key)
create new variables with the year added to the column name

reshape wide coffee maize, i(country) j(year)
convert a long dataset to wide

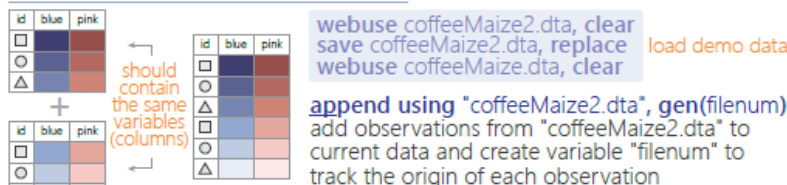
xpose, clear varname

transpose rows and columns of data, clearing the data and saving old column names as a new variable called "_varname"

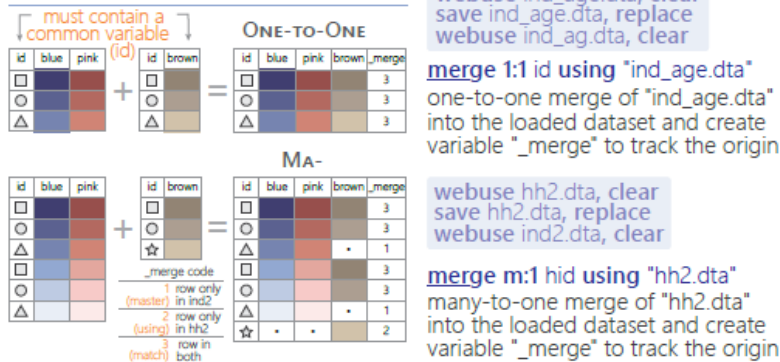
TIDY DATASETS have each observation in its own row and each variable in its own column.

Combine Data

ADDING (APPENDING) NEW DATA



MERGING TWO DATASETS TOGETHER



FUZZY MATCHING: COMBINING TWO DATASETS WITHOUT A COMMON ID

- reclink** match records from different data sets using probabilistic matching *ssc install reclink*
- jarowinkler** create distance measure for similarity between two strings *ssc install jarowinkler*

Manipulate Strings

GET STRING PROPERTIES

display length("This string has 29 characters")
return the length of the string

charlist make ** user-defined package*
display the set of unique characters within a string

display strpos("Stata", "a")
return the position in Stata where a is first found

FIND MATCHING STRINGS

display strmatch("123.89", "1???.9")
return true (1) or false (0) if string matches pattern

display substr("Stata", 3, 5)
return string of 5 characters starting with position 3

list make if regexm(make, "[0-9]*")
list observations where make matches the regular expression (here, records that contain a number)

list if regexm(make, "(Cad.|Chev.|Datsun)")
return all observations where make contains "Cad.", "Chev." or "Datsun"

list if inlist(word(make, 1), "Cad.", "Chev.", "Datsun")
compare the given list against the first word in make
return all observations where the first word of the make variable contains the listed words

TRANSFORM STRINGS

display regexr("My string", "My", "Your")
replace string1 ("My") with string2 ("Your")

replace make = subinstr(make, "Cad.", "Cadillac", 1)
replace first occurrence of "Cad." with Cadillac in the make variable

display strtrim(" Too much Space")
replace consecutive spaces with a single space

display trim(" leading / trailing spaces ")
remove extra spaces before and after a string

display strlower("STATA should not be ALL-CAPS")
change string case; see also **strupper**, **strproper**

display strtoname("1Var name")
convert string to Stata-compatible variable name

display real("100")
convert string to a numeric or missing value

Save & Export Data

compress
compress data in memory

save "myData.dta", replace *Stata 12-compatible file*
saveold "myData.dta", replace version(12)
save data in Stata format, replacing the data if a file with same name exists

export excel "myData.xls", /*
***/ firstrow(variables) replace**
export data as an Excel file (.xls) with the variable names as the first row

export delimited "myData.csv", delimiter(",") replace
export data as a comma-delimited file (.csv)

Data Visualization with Stata 15

Cheat Sheet

For more info see Stata's reference manual (stata.com)

ONE VARIABLE

[sysuse](#) [auto](#), [clear](#)

CONTINUOUS



histogram mpg, width(5) freq **kdensity** kdenopts(bwidth(5))

histogram

bin(#) • width(#) • density • fraction • frequency • percent • addlabels
addlabopts(<options>) • normal • normopts(<options>) • kdensity
kdenopts(<options>)



kdensity mpg, bwidth(3)

smoothed histogram

bwidth • kernel(<options>)
normal • normopts(<line options>)

main plot-specific options;
see help for complete set

DISCRETE



graph bar (count), over(foreign, gap(*0.5)) intensity(*0.5)

bar plot

(axis) • (percent) • (count) • over(<variable>, <options>: gap(*#) •
relabel • descending • reverse) • cw • missing • nolil • allcategories •
percentages • stack • bargap(#) • intensity(*) • yalternate • xalternate



graph bar (percent), over(rep78) over(foreign)

grouped bar plot

(axis) • (percent) • (count) • over(<variable>, <options>: gap(*#) •
relabel • descending • reverse) • cw • missing • nolil • allcategories •
percentages • stack • bargap(#) • intensity(*) • yalternate • xalternate



graph hbar draws horizontal bar charts

DISCRETE X, CONTINUOUS Y



graph bar (median) price, over(foreign)

bar plot

(axis) • (percent) • (count) • (stat: mean median sum min max ...) •
over(<variable>, <options>: gap(*#) • relabel • descending • reverse
sort(<variable>)) • cw • missing • nolil • allcategories • percentages
stack • bargap(#) • intensity(*) • yalternate • xalternate



graph dot (mean) length headroom, over(foreign) m(1, ms(5))

dot plot

(axis) • (percent) • (count) • (stat: mean median sum min max ...) •
over(<variable>, <options>: gap(*#) • relabel • descending • reverse
sort(<variable>)) • cw • missing • nolil • allcategories • percentages
linegap(#) • marker(#, <options>) • linestyle(dot | line | rectangle)
dots(<options>) • lines(<options>) • rectangles(<options>) • rwidth



graph hbox mpg, over(rep78, descending) by(foreign) missing

box plot

over(<variable>, <options>: total • gap(*#) • relabel • descending • reverse
sort(<variable>)) • missing • allcategories • intensity(*) • boxgap(#) •
medtype(line | marker) • medline(<options>) • medmarker(<options>)



vioplot price, over(foreign)

violin plot

over(<variable>, <options>: total • missing) • nolil •
vertical • horizontal • obs • kernel(<options>) • bwidth(#) •
barwidth(#) • dscale(#) • ygap(#) • qgap(#) • density(<options>) •
bar(<options>) • median(<options>) • obsopts(<options>)

Plot Placement

JUXTAPOSE (FACET)



twoway scatter mpg price, by(foreign, norescale)

total • missing • colfirst • rows(#) • cols(#) • holes(<numlist>) •
compact • nolabeled • nolinescale • nolinescale • nolinescale • nolinescale
nolixaxes • nolixaxes • nolixtick • nolixtick • nolixlabel
nolixlabel • nolixtitle • nolixtitle • imargin(<options>)

SUPERIMPOSE



graph combine plot1.gph plot2.gph...

combine 2+ saved graphs into a single plot

scatter y3 y2 y1 x, msymbol(i o i) mlabel(var3 var2 var1)

plot several y values for a single x value

graph twoway scatter mpg price in 27/74 || scatter mpg price /*

*/ if mpg < 15 & price > 12000 in 27/74, mlabel(make) m(i)
combine twoway plots using ||

BASIC PLOT SYNTAX:

graph <plot type> $y_1 y_2 \dots y_n x$ [in] [if], <plot options> by(var) xline(xint) yline(yint) text(y x "annotation")

variables: y first
plot-specific options — facet — annotations
titles
axes
custom appearance
plot size
save
<marker, line, text, axis, legend, background options> scheme(s1mono) play(customTheme) xsize(5) ysize(4) saving("myPlot.gph", replace)

TWO+ CONTINUOUS VARIABLES



graph matrix mpg price weight, half
scatter plot of each combination of variables

half • jitter(#) • jitterseed(#) •
diagonal • [aweight(<variable>)]



twoway scatter mpg weight, jitter(7)

scatter plot

jitter(#) • jitterseed(#) • sort • cmissing(yes | no)
connect(<options>) • [aweight(<variable>)]



twoway scatter mpg weight, mlabel(mpg)

scatter plot with labelled values

jitter(#) • jitterseed(#) • sort • cmissing(yes | no)
connect(<options>) • [aweight(<variable>)]



twoway connected mpg price, sort(price)

scatter plot with connected lines and symbols

jitter(#) • jitterseed(#) • sort **see also line**
connect(<options>) • cmissing(yes | no)



twoway area mpg price, sort(price)

line plot with area shading

sort • cmissing(yes | no) • vertical • horizontal
base(#)



twoway bar price rep78

bar plot

vertical • horizontal • base(#) • barwidth(#)



twoway dot mpg rep78

dot plot

dcolor(<color>) • dffcolor(<color>) • dlabel(<color>) •
dsizer(<markersize>) • dsymbol(<marker type>) •
dwidth(<stroke size>) • dotextend(yes | no)



twoway dropline mpg price in 1/5

dropped line plot

vertical • horizontal • base(#)



twoway rcapsym length headroom price

range plot ($y_1 \div y_2$) with capped lines

vertical • horizontal **see also rcap**



twoway rarea length headroom price, sort

range plot ($y_1 \div y_2$) with area shading

vertical • horizontal • sort
cmissing(yes | no)



twoway rbar length headroom price

range plot ($y_1 \div y_2$) with bars

vertical • horizontal • barwidth(#) • mwidth
msize(<marker size>)



twoway pcspike wage68 ttl_exp68 wage88 ttl_exp88

Parallel coordinates plot

vertical • horizontal



twoway pccapsym wage68 ttl_exp68 wage88 ttl_exp88

Slope/bump plot

vertical • horizontal • headlabel

THREE VARIABLES



twoway contour mpg price weight, level(20) crule(intensity)

3D contour plot

cuts(#s) • levels(#s) • minmax • crule(hue | chue | intensity | linear) •
gcolor(<color>) • gcolor(<color>) • gcolor(<colorlist>) • heatmap
interp(thinplatespline | shepard | none)



regress price mpg trunk weight length turn, nocons

matrix regmat = e(V)

ssc install plotmatrix

plotmatrix, mat(regmat) color(green)

heatmap

mat(<variable>) • split(<options>) • color(<color>) • freq

SUMMARY PLOTS



twoway mband mpg weight || scatter mpg weight

plot median of the y values

bands(#)



binscatter weight mpg, line(none)

plot a single value (mean or median) for each x value

medians • nquantiles(#) • discrete • controls(<variables>) •
linetype(lfit | qfit | connect | none) • aweight(<variable>)

FITTING RESULTS



twoway lfitci mpg weight || scatter mpg weight

calculate and plot linear fit to data with confidence intervals

level(#) • stdp • stdf • nolil • fitplot(<plottype>) • ciplot(<plottype>) •
range(#) • n(#) • atobs • estopts(<options>) • preopts(<options>)



twoway lowess mpg weight || scatter mpg weight

calculate and plot lowess smoothing

bwidth(#) • mean • noweight • logit • adjust



twoway qftci mpg weight, alwidth(none) || scatter mpg weight

calculate and plot quadratic fit to data with confidence intervals

level(#) • stdp • stdf • nolil • fitplot(<plottype>) • ciplot(<plottype>) •
range(#) • n(#) • atobs • estopts(<options>) • preopts(<options>)

REGRESSION RESULTS



regress price mpg headroom trunk length turn

coefplot, drop_cons xline(0)

Plot regression coefficients

baselevels • b(<options>) • at(<options>) • noci • levels(#)
keep(<variables>) • drop(<variables>) • rename(<list>)
horizontal • vertical • generate(<variable>)



regress mpg weight length turn

margins, eyex(weight) at(weight = (1800(200)4800))

marginsplot, noci

Plot marginal effects of regression

horizontal • noci

